

Activity 3 — Fishbone (Ishikawa) Diagram for Breadth of Causes

Horizon Bank — branch service complaints at Jurong East

Topic	Topic 1
Objective	A2 · K2, K5 — Categorise potential causes across all contributing dimensions.
Duration	45 minutes
Grouping	Teams of 3-5
Tools	Fishbone ed-tool, ChatGPT / Copilot / Gemini
Ed-tool	Fishbone Tool — https://alfredang.github.io/fishbone/

The Scenario

Horizon Bank's Jurong East branch has moved from best to worst in the retail network on customer experience in eleven months. Complaints run at 60 per month against a network average of 14. Average counter wait time is 58 minutes at peak; the service standard is 15.

Head Office ran a 'Service Excellence' refresher for all branch staff four months ago. Complaints did not move. The Branch Manager is now under performance review and morale is visibly poor — two tellers have resigned this quarter.

The Regional Director suspects the training was aimed at the wrong cause. Your team has been asked to map ALL contributing causes across every dimension before another intervention is funded. 5 Whys gave depth on one branch; this problem is too broad for one branch.

The Evidence

Horizon Bank Jurong East — observation log and system data

Dimension	Observation
Queue	Single queue for all transaction types; no triage between simple and complex
Peak	11:30-14:00 accounts for 61% of daily footfall (lunch crowd from nearby offices)
Counters	6 counters exist; mean 3.2 staffed at peak
Systems	New KYC platform launched 11 months ago; teller must key client data into 3 systems
Systems	Mean core-banking screen response 8s, was 2s before the KYC platform
Staff	44% of tellers have under 6 months tenure
Staff	Teller KPI is transactions-per-hour; no customer-experience measure
Customer	34% of customers arrive without required documents
Customer	Branch signage and the app's document checklist do not match
Process	Complex cases (loans, disputes) occupy a counter

	for a mean of 27 minutes
Management	No structured feedback route from branch to Head Office product owners
Environment	Only 18 seats in the waiting area; standing customers visibly agitated

Discussion Questions

1. Populate all six bones. Which bone filled up fastest, and which was hardest? What does an empty bone usually mean — no causes, or no visibility?
2. Head Office's answer was a Service Excellence training refresher. Which bone does that intervention target, and what share of your evidenced causes sit on that bone?
3. Find a cause on your diagram that CREATES another cause on a different bone. (Hint: look at the KPI and the queue.) What does that tell you about treating bones as independent?
4. Which three causes would you investigate first? Defend the ranking on evidence strength and likely contribution — not on ease of fixing.
5. The teller KPI is transactions-per-hour. Predict the behaviour that KPI produces at a counter when a confused elderly customer needs 20 minutes. Which bone does that belong on?

Step-by-Step

6. Read the Horizon Bank scenario and the observation log.
7. Open the Fishbone ed-tool at <https://alfredang.github.io/fishbone/>
8. Enter the problem statement as the fish head, with the 60-complaints and 58-minute numbers.
9. Label the six bones: People, Process, Technology, Material/Information, Environment, Measurement.
10. Work bone by bone as a team. For each observation in the log, decide which bone it belongs on and phrase it as a CAUSE, not a complaint.
11. Mark every cause [EVIDENCED] if a number in the log supports it, or [HYPOTHESIS] if it is your team's inference.
12. Run the GenAI Fishbone prompt from the slide and add any bone entries your team missed.
13. Draw an arrow between any two causes on DIFFERENT bones where one drives the other. Note how many arrows you find.
14. Count the evidenced causes per bone. Identify which bone Head Office's training actually targeted.
15. Circle the three causes you would investigate first and write one line of justification for each.
16. Screenshot or export the diagram and present your top three plus your verdict on the training decision.

The GenAI Prompt

Act as a quality improvement facilitator running an Ishikawa (fishbone) analysis with a bank branch team.

Problem (the fish head): Customer complaints at the Horizon Bank Jurong East branch have risen to 60 per month (network average 14), with peak counter wait time at 58 minutes against a 15-minute standard.

Generate candidate causes under these six bones: People, Process, Technology, Material/Information, Environment, Measurement.

Rules:

- Give 4-6 candidate causes per bone, each phrased as a CAUSE not a complaint
- Mark each cause [EVIDENCED] or [HYPOTHESIS] based only on what I give you below
- Do NOT propose solutions
- Finish by naming the 3 causes you would investigate first and say why

Evidence available: single queue with no triage; 61% of footfall between 11:30-14:00; 3.2 of 6 counters staffed at peak; new KYC platform requires triple data entry; screen response degraded 2s to 8s; 44% of tellers under 6 months tenure; teller KPI is transactions-per-hour only; 34% of customers arrive without documents; app checklist and branch signage disagree; complex cases take 27 minutes at the counter; no branch-to-HQ feedback route; 18 seats in the waiting area.

Trainer Debrief

Expected answers and the points to draw out. Trainer reference.

EXPECTED DISTRIBUTION — a well-built diagram is heavily loaded on Process, Technology and Measurement, and only lightly on People:

PEOPLE — 44% of tellers under 6 months; two resignations; low morale; thin coaching.

PROCESS — single queue, no triage; complex cases (27 min) block counters; rostering does not match the 11:30-14:00 peak (3.2 of 6 counters staffed).

TECHNOLOGY — KYC platform forces triple data entry; screen response degraded 2s to 8s.

MATERIAL/INFORMATION — app checklist contradicts branch signage; 34% arrive without documents.

ENVIRONMENT — 18 seats for a 58-minute wait; standing customers escalate faster.

MEASUREMENT — transactions-per-hour KPI with no experience measure; no branch-to-HO feedback loop.

KEY TEACHING POINTS:

1. THE HEADLINE: Head Office trained the PEOPLE bone, which carries the fewest evidenced causes. That is precisely why complaints did not move and S\$ was wasted. Learners should leave able to say: 'we intervened on the wrong bone.'
2. CROSS-BONE CAUSATION: the transactions-per-hour KPI (Measurement) drives tellers to rush or avoid complex cases (Process), which pushes complex customers back into the queue (Process), which lengthens the wait (Environment), which raises complaint volume. Bones are NOT independent — this observation is the bridge into System Loops in Activity 5.
3. FISHBONE vs 5 WHYS: 5 Whys gave one deep chain; Fishbone gives six shallow ones. Depth without breadth means you fix one cause and the problem persists; breadth without depth means you fix symptoms everywhere. Mature practice uses both — and the deliberate contrast between Activity 2 and Activity 3 is the point of running them back to back.
4. GenAI is excellent at populating bones fast (breadth at near-zero cost) and poor at knowing which are real. Note how many causes it returned as [HYPOTHESIS] — those are the ones your team must go and verify. The AI generates; the human evidences.

Self-Check — Is Your Output Finished?

Your fishbone is complete when: every bone has at least three entries; every entry is marked [EVIDENCED] or [HYPOTHESIS]; every entry is phrased as a cause ('single queue with no triage') rather than a complaint ('the queue is terrible') or a solution ('add more counters'); and you have found at least one cross-bone arrow. If any bone is empty, you have a visibility gap, not an absence of causes — say so explicitly.